

```

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ gcc -o lab04_v3 lab04_v3.c -lpthread -lm
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ chmod +x table1
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ./lab04_v3 5001
Usage: ./lab04_v3 <n> <p> <s> <config_file> [input_file]core_id]
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ./lab04_v3 3 5001 0 1 config.txt
[Slave 1] Listening on port 5001...
[Slave 1] Connected by parent.
[Slave 1] Received 2 rows (rows 0 to 1), covers 1 slave(s)
[Slave 1] My submatrix (2 x 3, rows 0 to 1):
11 12 13
14 15 16
[Slave 1] Sent 'ack' to parent.
time elapsed: 0.000076 seconds
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ./lab04_v3 5002 0 1 config.txt
[Slave 2] Listening on port 5002...
[Slave 2] Connected by parent.
[Slave 2] Received 1 rows (rows 2 to 2), covers 1 slave(s)
[Slave 2] My submatrix (1 x 3, rows 2 to 2):
17 18 19
[Slave 2] Sent 'ack' to parent.
time elapsed: 0.000000 seconds
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$

```

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$./lab03_v3 5000 0 config.txt input.txt

bash: ./lab03_v3: No such file or directory

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$./lab04_v3 5000 0 config.txt input.txt

No slaves in config.

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$./lab04_v3 3 5000 0 config.txt input.txt

[Master] n=3, slaves=2, mode=file

--- Full Matrix M (3 x 3) ---

```

11 12 13
14 15 16
17 18 19

```

[Master] Slave 1 (127.0.0.1:5001) -> 2 rows (rows 0 to 1)

```

11 12 13
14 15 16

```

[Master] Slave 2 (127.0.0.1:5002) -> 1 rows (rows 2 to 2)

```

17 18 19

```

[Master] Tree: 2 direct children (O(log 2))

[Master] Received 'ack' from Slave 2 (subtree: 1)

[Master] Received 'ack' from Slave 1 (subtree: 1)

time elapsed: 0.000732 seconds

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$./table1.sh

Error: ./lab04_v2 not found.

```

Compile first: gcc -o lab04_v2 lab04_v2.c -lpthread -lm
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ gcc -o lab04_v3
lab04_v3.c -lpthread -lm
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ./table1.sh
Error: ./lab04_v2 not found.
Compile first: gcc -o lab04_v2 lab04_v2.c -lpthread -lm
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ sed -i
's/lab04_v2/lab04_v3/g' table1.sh table2.sh table3.sh
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ./table1.sh
=====

```

Table 1: Single PC, No Core Affinity

=====

```

[1/36] n=4000, t=2, run 1 ... 0.014540 seconds
[2/36] n=4000, t=2, run 2 ... 0.018172 seconds
[3/36] n=4000, t=2, run 3 ... 0.018506 seconds
[4/36] n=4000, t=4, run 1 ... 0.022927 seconds
[5/36] n=4000, t=4, run 2 ... 0.022501 seconds
[6/36] n=4000, t=4, run 3 ... 0.023492 seconds
[7/36] n=4000, t=8, run 1 ... 0.027694 seconds
[8/36] n=4000, t=8, run 2 ... 0.027366 seconds
[9/36] n=4000, t=8, run 3 ... 0.033009 seconds
[10/36] n=4000, t=16, run 1 ... 0.038022 seconds
[11/36] n=4000, t=16, run 2 ... 0.033686 seconds
[12/36] n=4000, t=16, run 3 ... 0.033495 seconds
[13/36] n=8000, t=2, run 1 ... 0.056518 seconds
[14/36] n=8000, t=2, run 2 ... 0.060112 seconds
[15/36] n=8000, t=2, run 3 ... 0.055731 seconds
[16/36] n=8000, t=4, run 1 ... 0.085382 seconds
[17/36] n=8000, t=4, run 2 ... 0.087451 seconds
[18/36] n=8000, t=4, run 3 ... 0.086881 seconds
[19/36] n=8000, t=8, run 1 ... 0.102653 seconds
[20/36] n=8000, t=8, run 2 ... 0.099831 seconds
[21/36] n=8000, t=8, run 3 ... 0.111726 seconds
[22/36] n=8000, t=16, run 1 ... 0.124436 seconds
[23/36] n=8000, t=16, run 2 ... 0.122716 seconds
[24/36] n=8000, t=16, run 3 ... 0.126551 seconds
[25/36] n=16000, t=2, run 1 ... 0.204243 seconds
[26/36] n=16000, t=2, run 2 ... 0.199779 seconds
[27/36] n=16000, t=2, run 3 ... 0.208155 seconds
[28/36] n=16000, t=4, run 1 ... 0.307341 seconds
[29/36] n=16000, t=4, run 2 ... 0.296739 seconds
[30/36] n=16000, t=4, run 3 ... 0.303263 seconds
[31/36] n=16000, t=8, run 1 ... 0.414202 seconds
[32/36] n=16000, t=8, run 2 ... 0.394432 seconds
[33/36] n=16000, t=8, run 3 ... 0.398659 seconds

```

[34/36] n=16000, t=16, run 1 ... 0.433494 seconds
 [35/36] n=16000, t=16, run 2 ... 0.459253 seconds
 [36/36] n=16000, t=16, run 3 ... 0.458044 seconds

=====
 Table 1 complete! Results saved to: lab04_table1.csv
 =====

n	t	Run 1	Run 2	Run 3	Average
4000	2	0.014540	0.018172	0.018506	.017072
4000	4	0.022927	0.022501	0.023492	.022973
4000	8	0.027694	0.027366	0.033009	.029356
4000	16	0.038022	0.033686	0.033495	.035067
8000	2	0.056518	0.060112	0.055731	.057453
8000	4	0.085382	0.087451	0.086881	.086571
8000	8	0.102653	0.099831	0.111726	.104736
8000	16	0.124436	0.122716	0.126551	.124567
16000	2	0.204243	0.199779	0.208155	.204059
16000	4	0.307341	0.296739	0.303263	.302447
16000	8	0.414202	0.394432	0.398659	.402431
16000	16	0.433494	0.459253	0.458044	.450263

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$./table2.sh

=====
 Table 2: Single PC, With Core Affinity

=====
 Detected 16 cores. Using cores 1-15 for slaves.

[1/36] n=4000, t=2, run 1 ... 0.016576 seconds
 [2/36] n=4000, t=2, run 2 ... 0.015553 seconds
 [3/36] n=4000, t=2, run 3 ... 0.015187 seconds
 [4/36] n=4000, t=4, run 1 ... 0.023021 seconds
 [5/36] n=4000, t=4, run 2 ... 0.024071 seconds
 [6/36] n=4000, t=4, run 3 ... 0.024345 seconds
 [7/36] n=4000, t=8, run 1 ... 0.029421 seconds
 [8/36] n=4000, t=8, run 2 ... 0.030629 seconds
 [9/36] n=4000, t=8, run 3 ... 0.029055 seconds
 [10/36] n=4000, t=16, run 1 ... 0.035423 seconds
 [11/36] n=4000, t=16, run 2 ... 0.034414 seconds
 [12/36] n=4000, t=16, run 3 ... 0.035923 seconds
 [13/36] n=8000, t=2, run 1 ... 0.052038 seconds
 [14/36] n=8000, t=2, run 2 ... 0.056003 seconds
 [15/36] n=8000, t=2, run 3 ... 0.056865 seconds

[16/36] n=8000, t=4, run 1 ... 0.088928 seconds
 [17/36] n=8000, t=4, run 2 ... 0.086846 seconds
 [18/36] n=8000, t=4, run 3 ... 0.085981 seconds
 [19/36] n=8000, t=8, run 1 ... 0.103423 seconds
 [20/36] n=8000, t=8, run 2 ... 0.101003 seconds
 [21/36] n=8000, t=8, run 3 ... 0.109725 seconds
 [22/36] n=8000, t=16, run 1 ... 0.130102 seconds
 [23/36] n=8000, t=16, run 2 ... 0.122868 seconds
 [24/36] n=8000, t=16, run 3 ... 0.128082 seconds
 [25/36] n=16000, t=2, run 1 ... 0.219621 seconds
 [26/36] n=16000, t=2, run 2 ... 0.215163 seconds
 [27/36] n=16000, t=2, run 3 ... 0.225091 seconds
 [28/36] n=16000, t=4, run 1 ... 0.343269 seconds
 [29/36] n=16000, t=4, run 2 ... 0.333158 seconds
 [30/36] n=16000, t=4, run 3 ... 0.332469 seconds
 [31/36] n=16000, t=8, run 1 ... 0.414651 seconds
 [32/36] n=16000, t=8, run 2 ... 0.385119 seconds
 [33/36] n=16000, t=8, run 3 ... 0.414849 seconds
 [34/36] n=16000, t=16, run 1 ... 0.461068 seconds
 [35/36] n=16000, t=16, run 2 ... 0.494783 seconds
 [36/36] n=16000, t=16, run 3 ... 0.489821 seconds

=====
 Table 2 complete! Results saved to: lab04_table2.csv
 =====

n	t	Run 1	Run 2	Run 3	Average
4000	2	0.016576	0.015553	0.015187	.015772
4000	4	0.023021	0.024071	0.024345	.023812
4000	8	0.029421	0.030629	0.029055	.029701
4000	16	0.035423	0.034414	0.035923	.035253
8000	2	0.052038	0.056003	0.056865	.054968
8000	4	0.088928	0.086846	0.085981	.087251
8000	8	0.103423	0.101003	0.109725	.104717
8000	16	0.130102	0.122868	0.128082	.127017
16000	2	0.219621	0.215163	0.225091	.219958
16000	4	0.343269	0.333158	0.332469	.336298
16000	8	0.414651	0.385119	0.414849	.404873
16000	16	0.461068	0.494783	0.489821	.481890

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180\$ scp lab04)v3.c
 table3.sh cpdelosreyes@overqueen/home/cpdelosreyes
 bash: syntax error near unexpected token `)'

```

acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@overqueen/home/cpdelosreyes
cpdelosreyes@overqueen/home/cpdelosreyes: No such file or directory
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@10.0.9.20:~
cpdelosreyes@10.0.9.20's password:
Connection closed by 10.0.9.20 port 22
scp: Connection closed
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ scp lab04_v3.c
table3.sh cpdelosreyes@10.0.9.20:~
cpdelosreyes@10.0.9.20's password:
lab04_v3.c
100% 13KB 1.8MB/s 00:00
table3.sh
100% 6749 685.2KB/s 00:00
acer@acer-Veriton-X2722G:~/Downloads/CMSC 180_modified/CMSC 180$ ssh
cpdelosreyes@10.0.9.125
The authenticity of host '10.0.9.125 (10.0.9.125)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMlrEgLOjLE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.9.125' (ED25519) to the list of known hosts.
cpdelosreyes@10.0.9.125's password:
cpdelosreyes@fanny:~$ cd ~
cpdelosreyes@fanny:~$ gcc -o lab04_v3 lab04_v3.c -lpthread -lm
cpdelosreyes@fanny:~$ chmod +x lab04_v3 table3.sh
cpdelosreyes@fanny:~$ ssh-keygen -t rsa -N ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/cpdelosreyes/.ssh/id_rsa):
Created directory '/home/cpdelosreyes/.ssh'.
Your identification has been saved in /home/cpdelosreyes/.ssh/id_rsa
Your public key has been saved in /home/cpdelosreyes/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:bwFiFd4DnoYbv81VkxE8bfEsKFtwpYiODSZy5n+Zx8
cpdelosreyes@fanny.swarm.ics.uplb.edu.ph
The key's randomart image is:
+---[RSA 3072]----+
|  o+. o+.o  |
| .o==.. ..= = |
| ..==Bo oo* = + |
| =.*..o ooo o . |
| . + . S. . |
|  o .+o  |
|  o o..E  |
|  o o. .  |
|  o ..  |

```

```
+----[SHA256]-----+
cpdelosreyes@fanny:~$ ssh-copy-id cpdelosreyes@10.0.9.168
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
"/home/cpdelosreyes/.ssh/id_rsa.pub"
The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
The authenticity of host '10.0.9.168 (10.0.9.168)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
install the new keys
cpdelosreyes@10.0.9.168's password:

5f6b8e5a151da65450d3ec559b0511cc
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'cpdelosreyes@10.0.9.168'"
and check to make sure that only the key(s) you wanted were added.

```
cpdelosreyes@fanny:~$
cpdelosreyes@fanny:~$ 5f6b8e5a151da65450d3ec559b0511cc
-bash: 5f6b8e5a151da65450d3ec559b0511cc: command not found
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.168 "echo hello"
hello
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.158 "echo hello"
The authenticity of host '10.0.9.158 (10.0.9.158)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: 10.0.9.168
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.9.158' (ED25519) to the list of known hosts.
hello
cpdelosreyes@fanny:~$ ssh cpdelosreyes@10.0.9.125
The authenticity of host '10.0.9.125 (10.0.9.125)' can't be established.
ED25519 key fingerprint is SHA256:vl7W8nM3tEM5su+LVjapkA7f9DuER2O7CMIrEgLOjLE.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: 10.0.9.168
  ~/.ssh/known_hosts:4: 10.0.9.158
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.9.125' (ED25519) to the list of known hosts.
```

Last login: Thu Apr 16 15:27:12 2026 from 10.0.4.69
cpdelosreyes@fanny:~\$ cd ~
cpdelosreyes@fanny:~\$./table3.sh
Error: /home/cadelosreyes/lab04_v3 not found on this drone.
Upload to overqueen first: scp lab04_v3.c cadelosreyes@10.0.9.19:~/
Then compile: gcc -o lab04_v3 lab04_v3.c -lpthread -lm
cpdelosreyes@fanny:~\$